

Assignment 2 : The 10-Minute ML Explainer

The Goal

Work with a partner to explain **one** specific Machine Learning term. You need to show that you understand the theory and can explain it using a real-world example.

The Task

Prepare a **10-minute presentation** (5 minutes per person) divided into two parts or choose one of those to focus on it.

1. **The Theory:** What is the concept and why is it a problem?
2. **The Example:** Show a real scenario where this happens and how to fix it.

Choose a topic that interests you both: (I suggest a list below)

How You Will Be Graded

- **Simplicity:** Could another student understand this easily?
- **Accuracy:** Is the technical explanation correct?
- **Visuals:** Did you use clear drawings or charts instead of long sentences?
- **Teamwork:** Did both partners speak and connect their parts?

Check for Understanding (Before you start):

To make sure you are ready, ask yourselves these two questions:

1. **The "So What?" Test:** If this ML problem happens, what is the actual damage to the project?
2. **The "Analogy" Test:** Can you explain this concept to a non-engineer using a simple analogy (e.g., Overfitting is like a student memorizing a practice test but failing the real exam)?

Due Date: list of dates is below

Format: Live presentation

Suggested list :

1. Training Dynamics & Model Behavior (Understanding the "Fit")

1. **Overfitting:** Memorizing the noise vs. learning the pattern.
2. **Underfitting:** Why a model might be "too simple" for the data.
3. **The Bias-Variance Tradeoff:** Balancing flexibility with stability.
4. **Early Stopping:** Finding the perfect moment to stop training.
5. **Regularization (L1/L2):** Adding a "penalty" to prevent the model from over-complicating.
6. **Data Leakage:** When the model "cheats" by seeing the test answers early.
7. **Hyperparameter Tuning:** The difference between "learning" and "configuring" a model.

2. Niche Metrics (Beyond "Accuracy")

8. **Precision vs. Recall:** The cost of a False Alarm vs. a Missed Event.
9. **F1-Score:** The best metric for when your data is unbalanced.
10. **Confusion Matrix:** A visual map of exactly where the model gets confused.
11. **Log Loss:** Measuring how "confident" (and how wrong) a model's prediction is.
12. **Inference Latency:** Measuring speed—when a slow model is a useless model.
13. **False Positives vs. False Negatives:** Real-world consequences in medicine vs. security.

3. MLOps & Real-World Operations

14. **Model Drift:** What happens when the world changes but your model doesn't.
15. **Data Drift:** Detecting when the incoming data looks different from training data.
16. **Training-Serving Skew:** When the model behaves differently in the lab vs. in production.
17. **Model Monitoring:** How to build an "alarm system" for AI.
18. **A/B Testing for AI:** Comparing two models on real users safely.
19. **Feedback Loops:** When a model's own mistakes pollute its future training data.
20. **Feature Stores:** Keeping a consistent "library" of data for all your models.

4. Data Challenges & Handling

21. **Imbalanced Data:** Training a model when the "interesting" event is 0.1% of the data.
22. **Data Augmentation:** Creating "artificial" variety to make the model more robust.
23. **Outlier Detection:** Deciding which data points are "special" and which are "errors."
24. **Label Noise:** How to train a model when your human-labeled data is messy.
25. **Cross-Validation:** How to prove your model's success isn't just a "lucky" accident.

5. Advanced Niche Concepts (Simplified)

26. **Transfer Learning:** Taking a "pre-trained" brain and giving it a new job.
27. **Explainable AI (XAI):** Asking the "Black Box" to show its work.

28. **Quantization:** Shrinking a model to fit on a small device (like a phone).
29. **Retrieval-Augmented Generation (RAG):** Giving a model a "knowledge base" to prevent hallucinations.
30. **Adversarial Attacks:** Small, invisible changes that trick a model into making a mistake.